Missing Coin Sum

Time limit: 1.00 s
Memory limit: 512 MB

CSES 2183 | Sorting and Searching | https://cses.fi/problemset/task/2183

You have n coins with positive integer values. What is the smallest sum you cannot create using a subset of the coins?

Input

The first line has an integer n: the number of coins.

The second line has n integers $x_1,x_2,\ldots,x_n$: the value of each coin.

Output

Print one integer: the smallest coin sum.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le x_i \le 10^9$

Example

Input
5 2 9 1 2 7

Output
6